

InterviewIQ

Multi-Agent RAG Placement Preparation Companion

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Domain: Computer Science & Engineering (Generative AI)

Orchestration Framework: Langflow Engine (Local via UV Pipeline)

Core Reasoning Engine: IBM watsonx.ai (Granite-3.2-8b)

Vector Database Fabric: DataStax Astra DB (Serverless Vector)

Deployment Environment: Sydney Region Cloud Mesh (au-syd)

1. Executive Summary & Overview

InterviewIQ is an enterprise-grade multi-agent conversational simulation platform engineered to transform the traditional paradigm of campus recruitment preparation. Designed specifically to meet modern industrial standards, the architecture addresses critical bottlenecks in student mock-interviews by replacing static, generic question lists with a dynamic, context-aware artificial intelligence recruiter framework.

The backend engine utilizes advanced Retrieval-Augmented Generation (RAG) tied into high-performance vector databases. By offloading complex contextual alignment tasks onto distributed specialized nodes, the platform acts as both a rigorous technical interviewer and an encouraging soft-skills mentor. The final interaction compiles structured markdown evaluation report cards detailing precise performance matrices.

2. Technical Architecture Blueprint

The infrastructure topology is constructed entirely using an asynchronous decoupled sequence of graphical execution nodes. This architecture ensures computational efficiency and absolute data isolating guardrails:

- **Frontend Chat Boundary Node:** Intercepts human text prompts via a standardized local loopback socket inside the Langflow environment.
- **Orchestration Graph Engine:** Assembles logic workflows on a clean local workspace canvas, streaming prompt parameters efficiently to secure remote clusters.
- **IBM watsonx.ai Compute Cluster:** Houses the core foundation model weights, performing sophisticated natural language processing and clinical token sequence deduction.
- **DataStax Astra DB Core:** Stores indexed corporate technical questions and answer validation matrices inside high-performance serverless vector tables.

2.1 Core Foundational Models

The model processing core utilizes the **ibm/granite-3-2-8b** instruct pipeline. This specialized model architecture was selected due to its exceptional performance benchmarks in programming language logic parsing, technical framework evaluation (such as React Virtual DOM architectures, state lifecycle dynamics, and CSS styling layouts), and structured systemic tool-calling coordination.

3. Comprehensive Implementation Guide

This section lists the exact parameter controls, endpoints, and step-by-step pipeline configurations utilized to activate the deployment framework from absolute scratch.

3.1 Enterprise Cloud Configuration Paths

1. Access the global **IBM Cloud Console** hub and provision dedicated instances of *watsonx.ai Studio* and *watsonx.ai Runtime* inside the **Sydney (au-syd)** region matrix.
2. Isolate the target active orchestration region cluster endpoint url string:

```
https://au-syd.ml.cloud.ibm.com
```

3. Navigate to the cloud account's **Access (IAM)** directory matrix, generate a long-lived identity access verification credential token named InterviewIQ-Security-Token, and backup the generated unique key string securely.

3.2 Local Pipeline Workspace Activation

The orchestration container environment is loaded locally over the terminal interface using the highly optimized ultra-fast pipeline runtime manager:

```
uv run langflow run
```

Upon active initialization routines compiling completely, direct any standard browser window tab to the local loopback hosting address: `http://127.0.0.1:7860` to load the blank grid canvas area.

4. Multi-Agent Prompt System Constraints

The autonomous persona, step-by-step role tracking, and compliance metrics governing the AI recruiter node are injected directly via a comprehensive system instruction prompt block:

1. Interaction Behavior

- The agent must respond in a highly encouraging, professional, and mentoring tone.
- Always begin the session with a warm welcome to the corporate mock simulation.
- Maintain a realistic conversational interview flow instead of displaying all questions at once.
- Ask exactly one question at a time to allow the student to type their answer naturally.
- Acknowledge their response with constructive validation before asking the next question.

2. Data Collection Behavior

- Collect user preferences step-by-step: target job role (Frontend Developer) and target experience level (Fresher).
- Once fields are collected, notify the user that you are preparing their custom question matrix.

3. Decision-Making & Evaluation Behavior

- Query the reference data to present targeted core technical questions and HR behavioral scenarios.
- Compare their reply structure with professional industry parameters.

4. Response Behavior

- Provide a clean, structured scorecard at the end of the interview session.
- Output format: Candidate Summary, Score Out of 10, Areas of Strength, and Model Answer Key Tips.

5. Text Formatting Rule

- Do NOT output prompt templates, internal tags, or raw systemic code like "### Instruction:".
- Stop generating text immediately once you have provided your natural conversational answer.

5. Summary of Novelty & Core Uniqueness

Unlike simple boilerplate chatbots that loop over static prompt arrays, **InterviewIQ** establishes several clear vectors of modern technological novelty:

- **Decoupled Sequential Architecture:** Separates user target extraction, dataset retrieval, and assessment workflows into distinct functional roles, mitigating token degradation.
- **Dynamic Context Scaling:** Adjusts immediate next-turn question focus parameters based on semantic analytics score gaps extracted directly from user responses.

- **Zero-Hallucination Grounding:** Binds model reference operations entirely to verified evaluation datasets, preserving strict academic alignment tracking.
- **Centralized Cloud Governance:** Preserves end-to-end data privacy by utilizing industrial enterprise security guardrails native to the **IBM watsonx.ai** platform.